# Pankaj Kumar



Software Engineer Intern | Game Development | C# / Unity | VR
+91 7451042661 | pankajkumar_240666@aitpune.edu.in
LinkedIn | GitHub

## EDUCATION

**Army Institute of Technology, Pune** — **Pune, Maharashtra**
B.E. in Electronics & Telecommunication Engineering — 2024 – 2028

- Aggregate CGPA (Semesters I–IV): **7.82/10**

## TECHNICAL SKILLS

| | |
|---|---|
| **Programming** | C#, Java, Python, TypeScript, JavaScript |
| **Game Development** | Unity 6, XR Interaction Toolkit, OpenXR, XR Hands, Auto Hand, URP, Blender, Mixamo |
| **Game Systems** | Gameplay Programming, State Machines, Collision Detection, Physics, Ragdoll Systems, 3D Interaction, VR Development |
| **CS Fundamentals** | Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS, Complexity Analysis |
| **Backend** | Spring Boot, FastAPI, REST APIs, JWT Authentication, SQL, MySQL, PostgreSQL, SQLite, MongoDB |
| **AI / Tools** | Gemini, Google GenAI, AI Agents, Git, GitHub, Docker, Linux, Visual Studio, VS Code, Android Studio |

## EXPERIENCE

**AR/VR Intern** — **Remote**
IDZ Digital Private Limited — Aug 2025 – Jan 2026

- Developed interactive AR/VR features using **C# and Unity3D**, working with real-time 3D interactions and immersive application systems.
- Worked with 3D assets and interactive components to build and refine immersive application experiences.

**VR Lead – GDXR Club** — **Pune**
Army Institute of Technology — Aug 2024 – Present

- Led AR/VR development using **Unity, Blender and Mixamo**, contributing to interactive 3D projects and coordinating technical development.
- Worked on 3D workflows, asset preparation and interactive VR experiences for student projects.

## PROJECTS

**FarmMarketVR – VR Farm-to-Shop Simulator**

- Built a Unity 6 VR simulation featuring farming, harvesting, inventory, shelf stocking, pricing, checkout and an in-game economy using **C#, OpenXR, XR Interaction Toolkit and Auto Hand**.
- Implemented a **customer behavior state machine** covering product search, shelf navigation, purchasing, queues, restocking reactions, customer avoidance and post-payment reactions.
- Developed gameplay systems for land unlocking, product data, stock placement, furniture interaction, checkout and VR/desktop controls; supports up to **10 concurrent customers**.
- GitHub

**RCoreEngine – C# Game Engine**

- Developed a C# game-engine project exploring reusable engine architecture, real-time game programming and object-oriented software design.
- Structured engine components around modular and reusable systems while applying core programming and software-engineering principles.
- GitHub

**NPC Punch Hit VR – Physics-Based VR Combat Prototype**

- Developed a Unity VR combat prototype using **C#, Auto Hand and physics-based interactions** for glove, projectile and melee collision systems.
- Implemented directional hit detection across **head, chest, back and leg zones**, spawning contact-point VFX and triggering context-specific reactions.
- Built impact physics using contact-point impulses, active-ragdoll reactions and per-zone cooldown management to coordinate hit animations.
- GitHub

**FinLens – AI Financial Intelligence Platform**

- Engineered a full-stack financial document intelligence platform using **React, FastAPI, Python, SQLite and Docker**, with authenticated REST APIs and multimodal Gemini processing.
- Implemented OCR-based document extraction and AI-assisted cross-document financial analysis using multimodal models. GitHub

**Recover**$_{AI}$ $--AIRecovery/IntelligencePlatform$

- **Developed an AI-focused software project exploring intelligent automation and practical AI-assisted**

workflows.
- Applied modern software-development practices to build and structure the project for reusable, maintainable functionality.
- GitHub

Media-Automation – Automated Media Processing Platform
- Developed an automation-focused project for streamlining media-processing workflows and reducing repetitive manual tasks.
- Structured the project around reusable automation components and practical software-engineering workflows.
- GitHub

LedgerZero – Financial / Ledger Platform
- Developed a financial software project focused on ledger-oriented workflows and structured financial data.
- Applied backend and software-engineering concepts to organize financial operations into a maintainable application structure.
- GitHub

## GAME DEVELOPMENT PORTFOLIO

- Complete Games Compilation – Collection of game projects, Unity/VR prototypes, gameplay experiments and interactive 3D experiences developed throughout my game-development journey. View Portfolio

## ACHIEVEMENTS

- First Runner-up – Game Jam organized by IIIT Delhi in collaboration with IGDA.
- 200+ LeetCode problems solved across Data Structures, Algorithms and competitive problem solving.
- Finalist – Paytm AI Hackathon.
- Second Runner-up – Arcade, an AR-based lens-making competition.
- Finalist – Innerve 9, Vinhack at VIT Vellore, and Imaginate at Solution 2K25.